Anubis Quartz Industrial Research playbook

Follow the manual or Copilot-assisted route to calculate workload GPU requirements, prepare a design handoff, obtain a plain-language explanation and save the completed engagement.

Reader key. **UI field** identifies a screen label. `Entered value` is literal content. **[Button]** identifies an action. ***Checked*** and ***blank*** identify settings.

Customer scenario

A manufacturer wants GPU-accelerated fluid simulation for turbine design. Engineers describe meshes, solver steps and simulation deadlines. No text model, retrieval, agent, media generation or model training is involved.

All added numbers are explicit fictional customer planning assumptions for a reproducible demonstration. They are not historical measurements or qualified hardware benchmarks. Dense text components use dense8b, one call/request, 2000 input and 500 output tokens unless overridden in the intake table.

Workflow input register

ID	Scope and exact field	Value	Basis
I01	Engagement **Customer workload name**	`Quartz Industrial Research manual playbook / Quartz Industrial Research assisted playbook`	Entered
I02	Engagement **Factory users**	`One organisation`	Selected
I03	Engagement **Do activities run at the same time?**	`Yes — overlapping demand`	Selected
I04	Fluid simulation **Component name**	`Fluid simulation`	Selected
I05	Fluid simulation **Application pattern**	`other`	Selected
I06	Fluid simulation **Compute location**	`In this factory`	Selected
I07	Fluid simulation **Relationship**	`Independent`	Selected
I08	Fluid simulation **Sizing method**	`Reference capacity for independent work`	Selected

ID	Scope and exact field	Value	Basis
I09	Fluid simulation **Peak work units per second**	0.000138889	Explicit planning assumption
I10	Fluid simulation **Work unit**	simulation	Explicit planning assumption
I11	Fluid simulation **GPUs per reference group**	8	Explicit planning assumption
I12	Fluid simulation **Required GPU memory in GB**	90	Explicit planning assumption
I13	Fluid simulation **H200 work units per second per group**	0.000069444	Explicit planning assumption
I14	Fluid simulation **B200 work units per second per group**	0.000092592	Explicit planning assumption
I15	Fluid simulation **B300 work units per second per group**	0.000138888	Explicit planning assumption
I16	Fluid simulation **Independent reference groups can serve this demand**	Checked	Explicit planning assumption
I17	Fluid simulation **I accept this reference capacity and its scope**	Checked	Explicit planning assumption
I18	Fluid simulation **Reference source and conditions**	Fictional consultant planning assumption for this demonstration, not measured benchmark evidence. Twelve independent CFD runs per 24h. Fixed solver/version, 20-million-cell mesh, 10,000 steps, FP64 mixed solver. Each job fits one eight-GPU group. Assumed group run times H200 4h, B200 3h, B300 2h. Rates conservatively rounded. No linear scaling of one larger mesh.	Explicit planning assumption
I19	Shared **Infrastructure deployment profile**	HGX B300 · Spectrum-X · Enterprise reference design with two network planes	Default or explicitly left blank
I20	Shared **Sensitivity envelope**	±30%	Default or explicitly left blank
I21	Shared **Use these planning assumptions**	Checked	Default or explicitly left blank

ID	Scope and exact field	Value	Basis
I22	Shared **Locally retained data in TB / Daily growth in GB**	`Blank`	Default or explicitly left blank
I23	Shared **Initial dataset to load in TB / Initial loading window in hours**	`Blank`	Default or explicitly left blank
I24	Shared **Ongoing incoming / outgoing data in GB per day**	`Blank`	Default or explicitly left blank
I25	Shared **Aggregate storage reads / writes in Gbps**	`Blank`	Default or explicitly left blank
I26	Shared **East-west payload per GPU in Gbps / Customer datacenter round-trip target in ms**	`Blank`	Default or explicitly left blank
I27	Shared **Usable link fraction / Initial loading overlap**	`0.7 / Yes`	Default or explicitly left blank
I28	Shared **Explanation connection**	`Manual: deterministic. Assisted: tested DeepSeek connection`	Default or explicitly left blank
I29	Shared **This saved result is public or fictional and may be sent to the selected provider**	`Checked only for this fictional assisted example`	Default or explicitly left blank

Remaining displayed engine defaults: BF16 weights and KV cache, context 8192 tokens, 32 concurrent sequences, 8 GB workspace, compute efficiency 0.35 and memory efficiency 0.50. Host provisions are 256 GB RAM, 3840 GB local disk and 16 CPU cores per workload GPU. The blank growth horizon uses 365 days. Reference sizing does not use text-token equations.

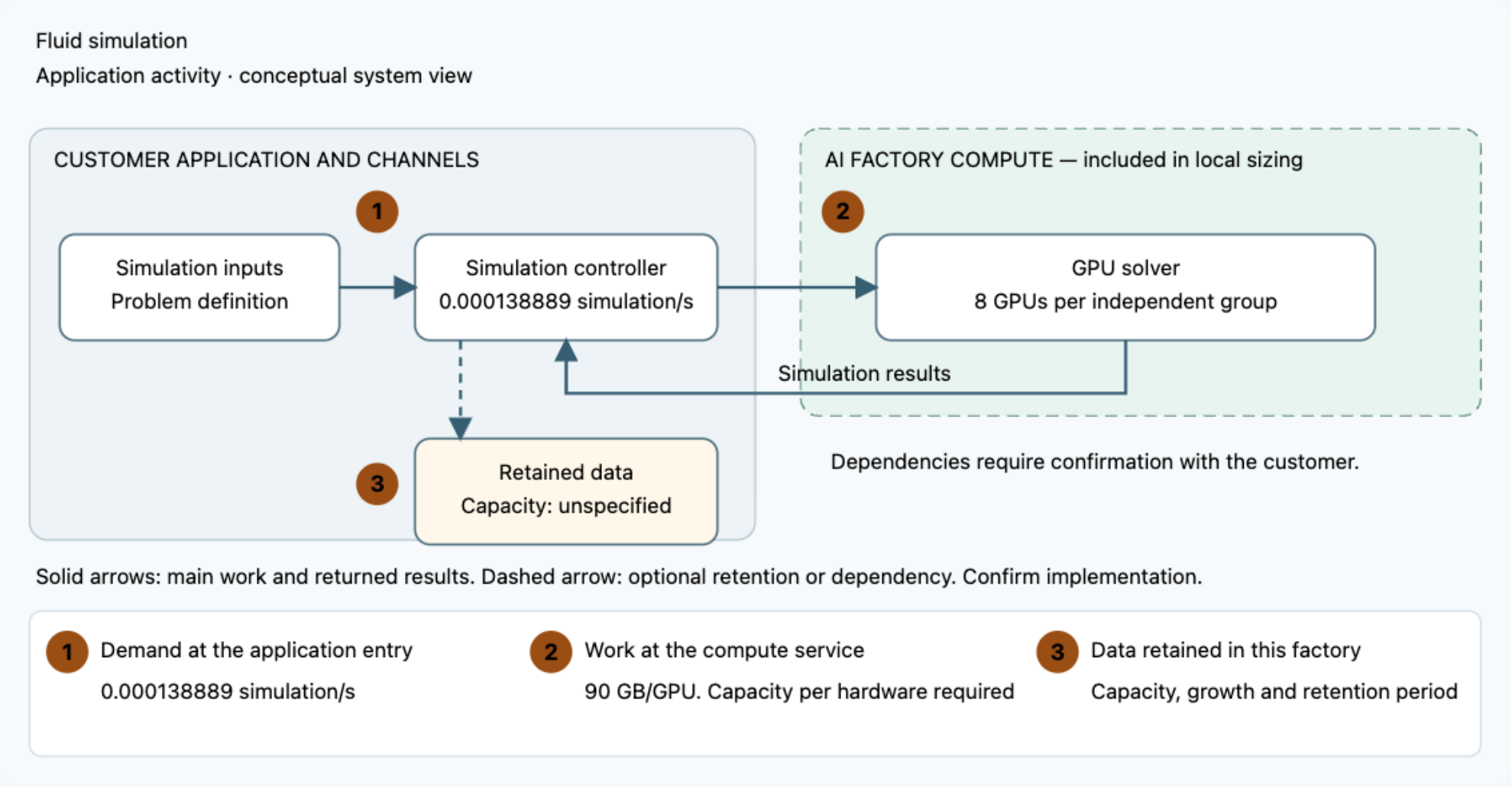
Expected result and process

B300 workload GPUs	B200 workload GPUs	H200 workload GPUs	Selected B300 factory anchor
16	16	24	256

Describe the application → confirm its boundary → enter demand and assumptions → calculate GPU comparisons → prepare design handoff → explain the saved result → complete and retrieve the engagement.

The workload count and factory anchor are separate. Spare compute may support additional work, but it does not establish a qualified tenant count. Outstanding network and shared-storage measurements remain visible in the companion requirements.

Fluid simulation architecture



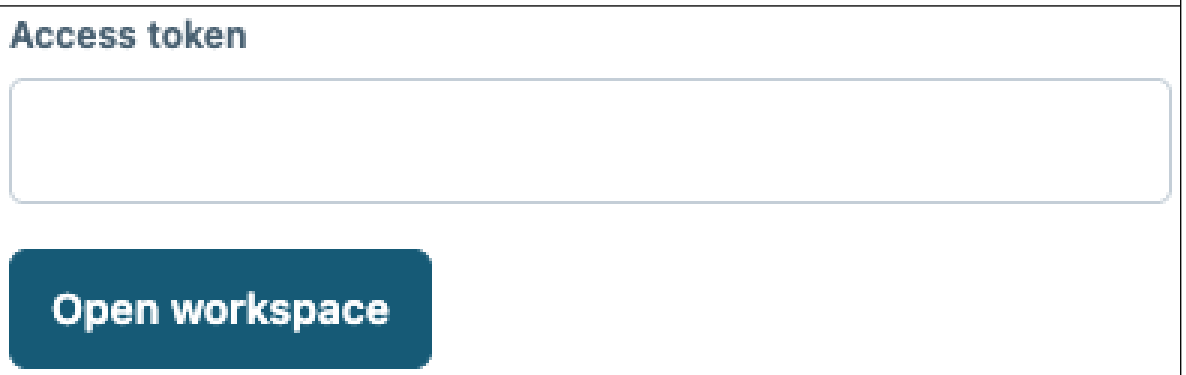
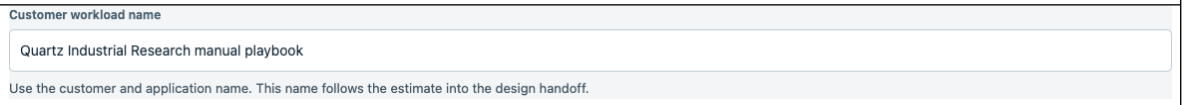
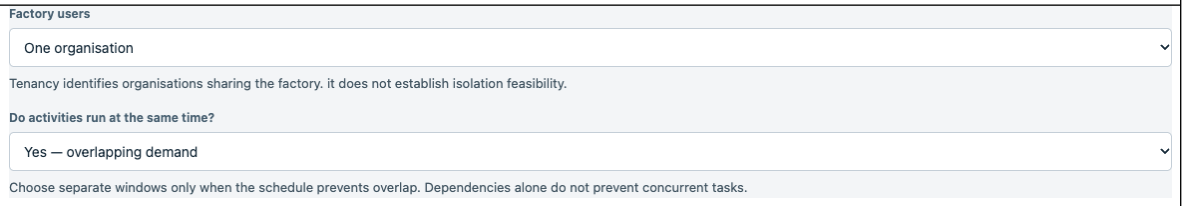
Start and access the workspace

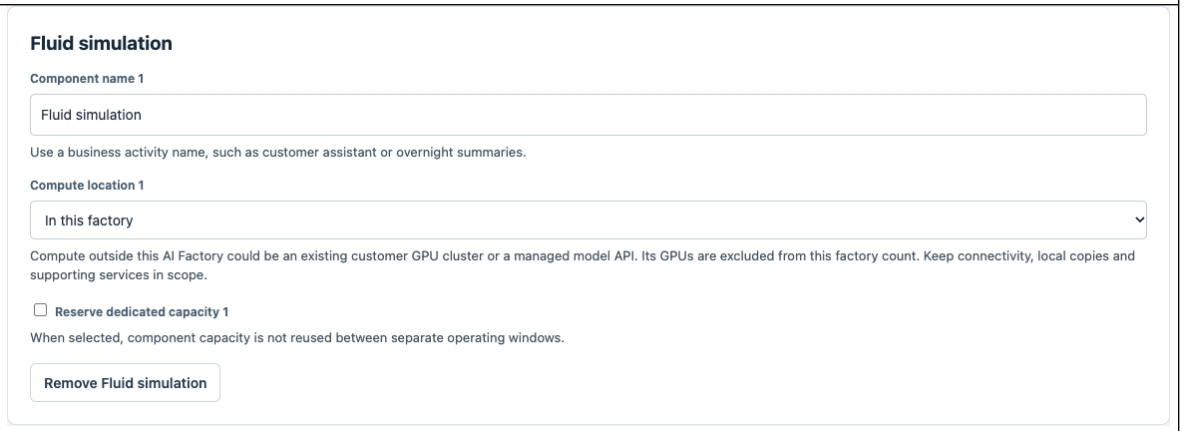
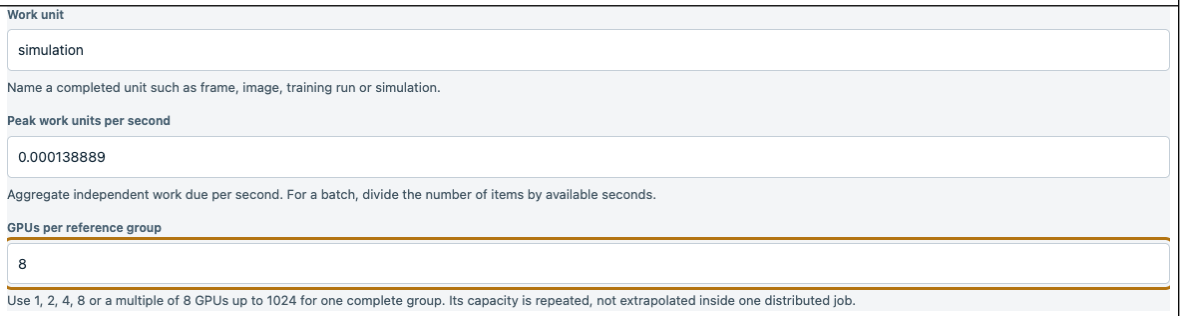
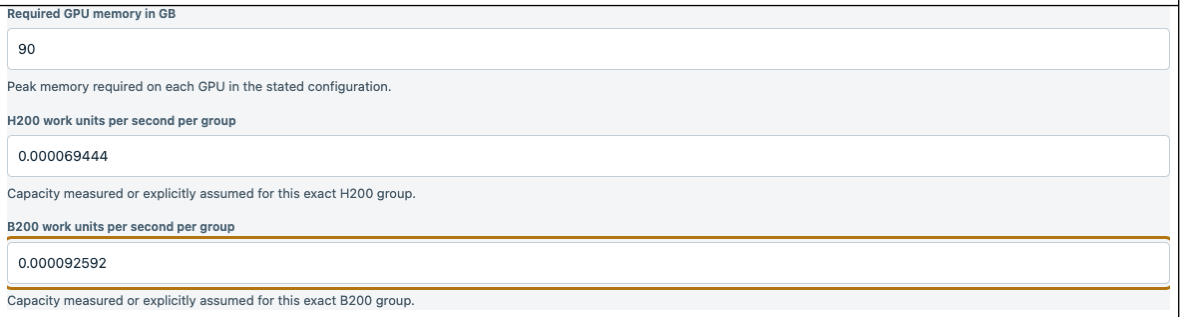
Complete the Mac installation guide in `docs/macos-installation-guide.txt`. From the repository folder run `python3 deploy/local/run.py`. Keep the terminal open. On first launch the application creates `.anubis/identities.json`. Open that private file locally, copy your identity token without quotation marks, and paste it into **Access token**. Click **[Open workspace]**. Saved cases persist in `.anubis/anubis.sqlite` across restarts. Preserve this directory and use the documented backup process.

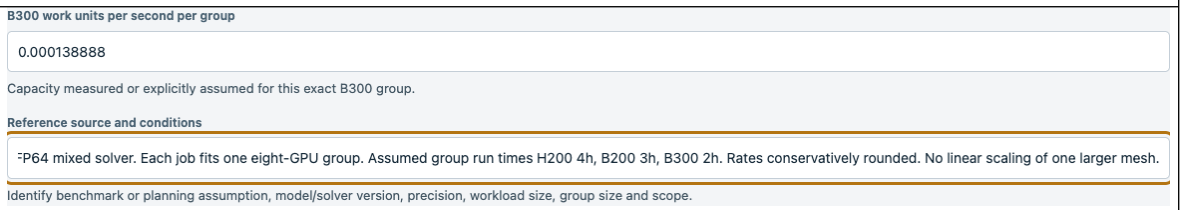
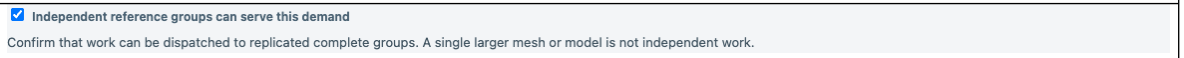
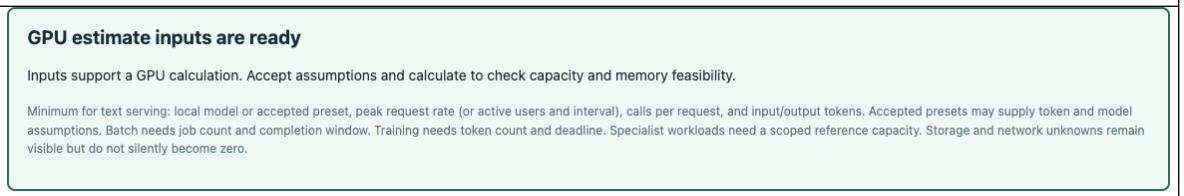
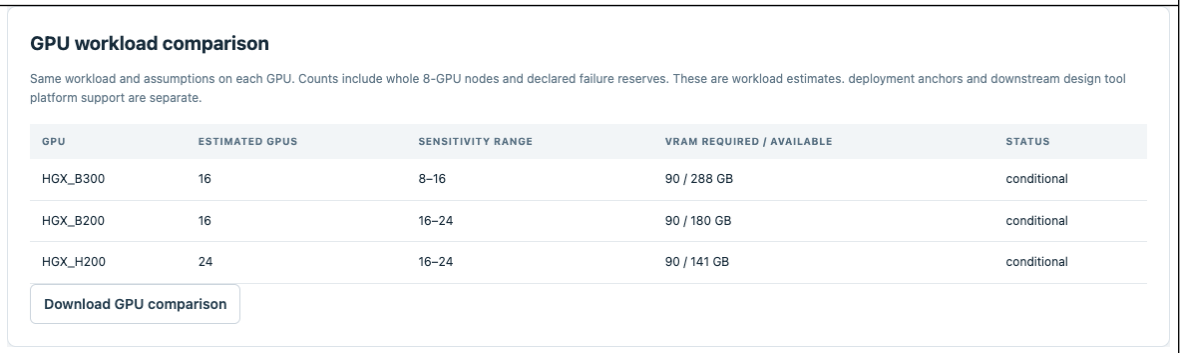
For Copilot, put the provider API key in the protected backend configuration, never in the chat or input register. Open **[AI connections]**, configure the approved provider and backend environment-variable name, and test the connection. The recorded assisted walkthrough uses DeepSeek deepseek-flash with low reasoning. Connection testing is separate from workload and explanation validation.

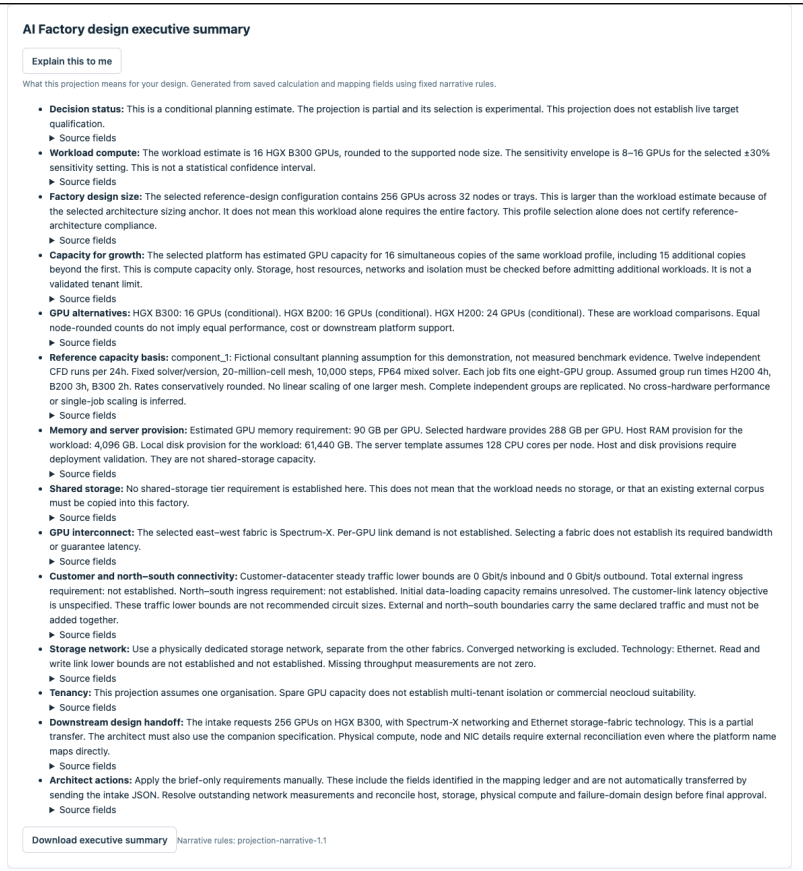
Manual workflow

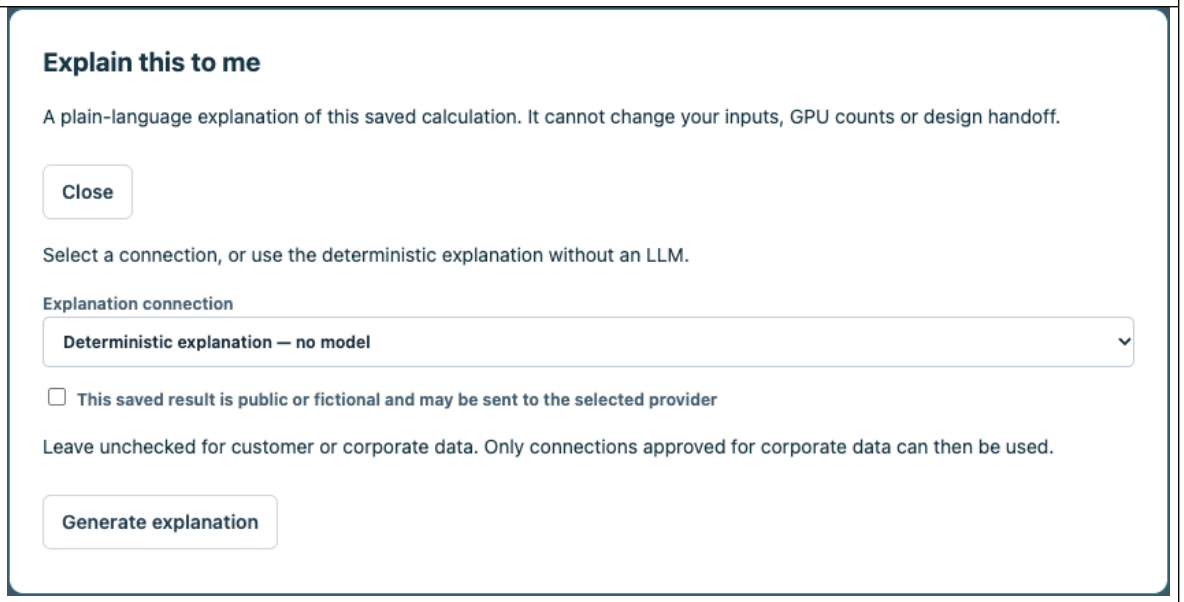
Follow these numbered steps in order. Use a separate engagement name for this route. Input values are the same workload assumptions in both routes. In the assisted route, inspect extracted facts and explicitly enter any specialist reference values in the guided form.

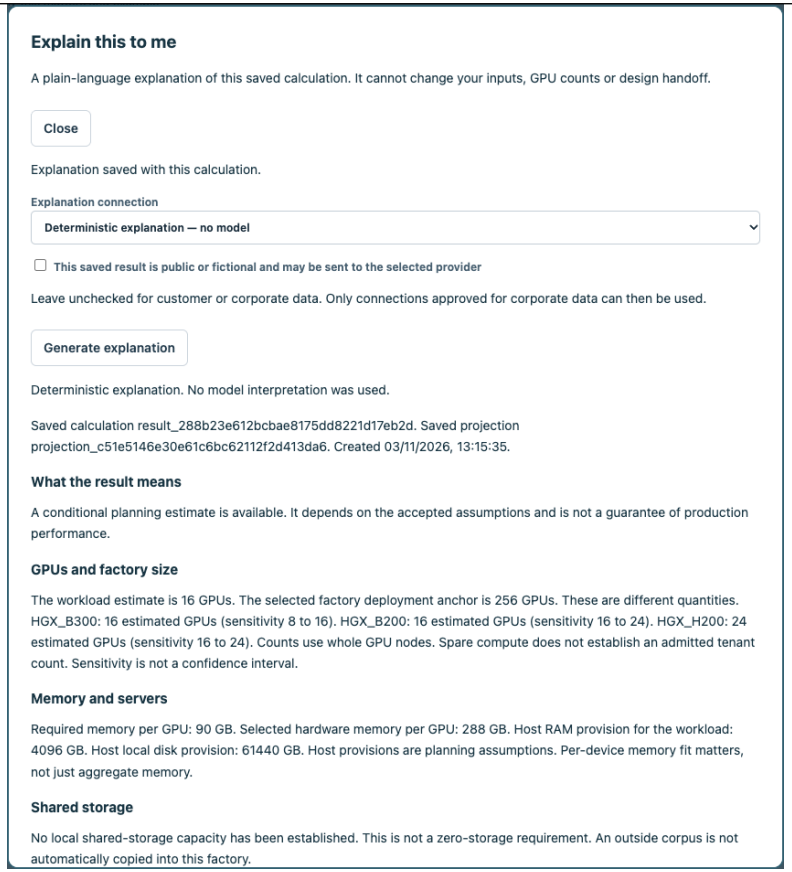
Step	Activity and exact values	Screen reference
1	Start Anubis and open the local address. Retrieve your private token from `.anubis/identities.json`. Paste it into **Access token** and click **[Open workspace]**. Never share the token.	
2	Click **[New engagement]**. Set **Customer workload name** = `Quartz Industrial Research manual playbook`. Click **[Add Other GPU application]**. Click **[Continue]**.	
3	Set **Factory users** = `One organisation`. Set **Do activities run at the same time?** = `Yes — overlapping demand`.	

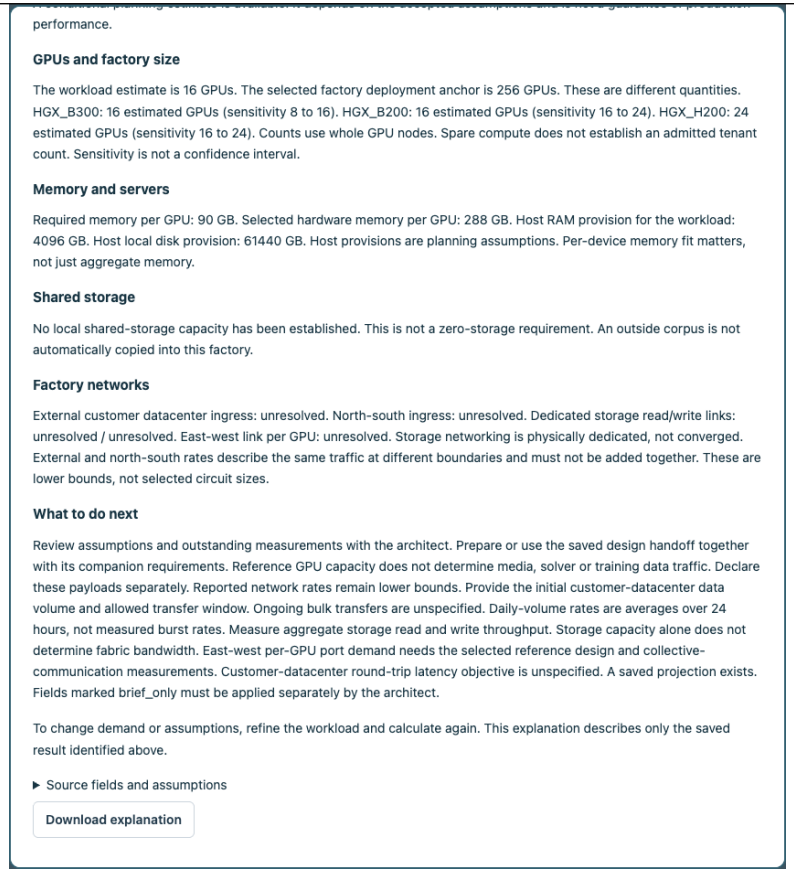
Step	Activity and exact values	Screen reference
4	Set **Component name 1** = `Fluid simulation`. Set **Compute location 1** = `In this factory`. Set **Reserve dedicated capacity 1** ***unchecked***.	
5	Select **[Fluid simulation]**. Set **Sizing method** = `Reference capacity for independent work`. This uses complete independent GPU groups and the explicit scenario assumptions.	
6	Set **Work unit** = `simulation`. Set **Peak work units per second** = `0.000138889`. Set **GPUs per reference group** = `8`.	
7	Set **Required GPU memory in GB** = `90`. Set **H200 work units per second per group** = `0.000069444`. Set **B200 work units per second per group** = `0.000092592`.	

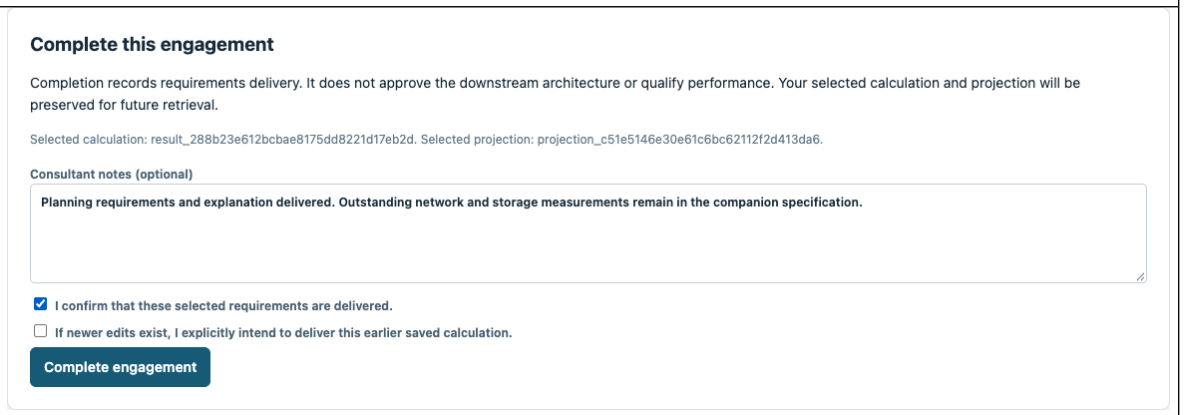
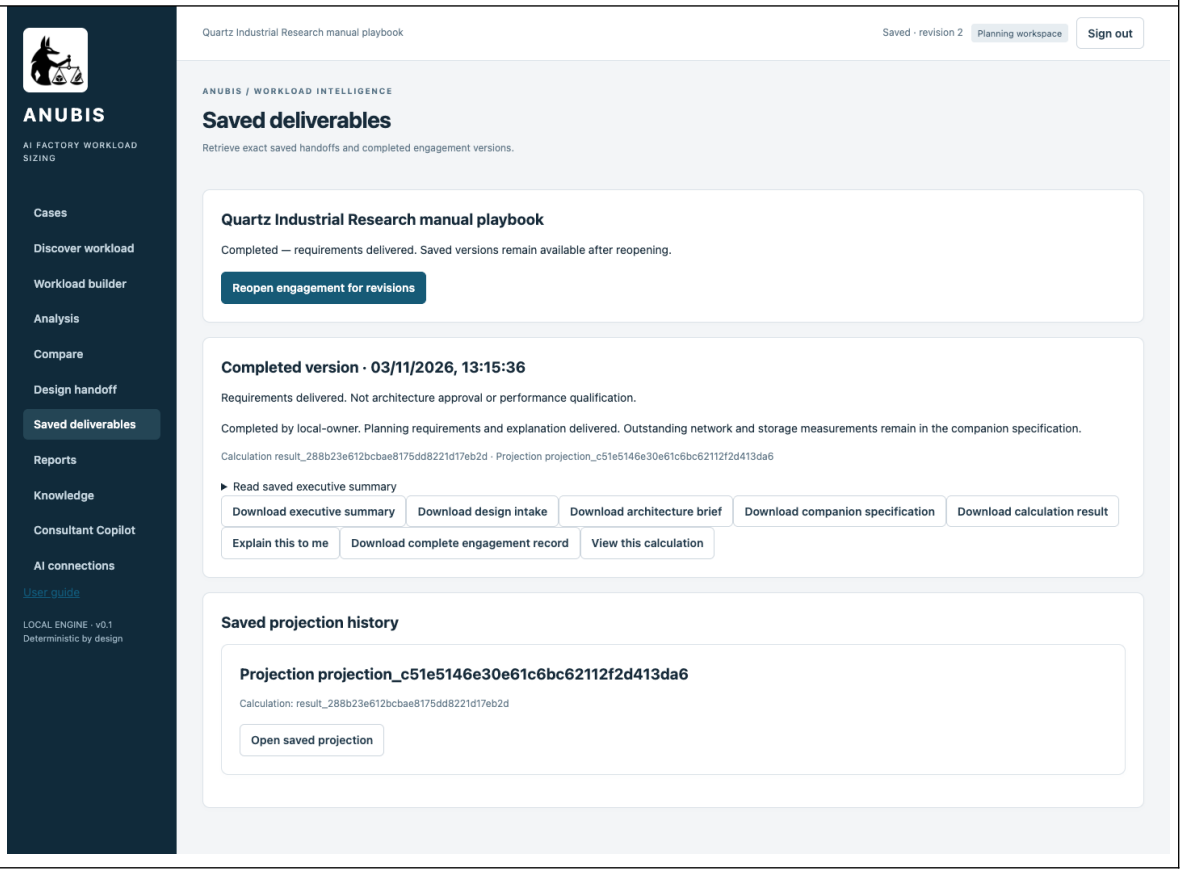
Step	Activity and exact values	Screen reference
8	Set **B300 work units per second per group** = `0.000138888`. Set **Reference source and conditions** = `Fictional consultant planning assumption for this demonstration, not measured benchmark evidence. Twelve independent CFD runs per 24h. Fixed solver/version, 20-million-cell mesh, 10,000 steps, FP64 mixed solver. Each job fits one eight-GPU group. Assumed group run times H200 4h, B200 3h, B300 2h. Rates conservatively rounded. No linear scaling of one larger mesh.`.	
9	Check **Independent reference groups can serve this demand** and **I accept this reference capacity and its scope**. Leave **Reference includes local search and indexing *unchecked*** for this component. These example capacities are accepted planning assumptions, not measured benchmark results.	
10	Confirm **[GPU estimate inputs are ready]**. Retain the displayed B300 enterprise deployment profile and **Sensitivity envelope** = `±30%`. Check **Use these planning assumptions**. Click **[Calculate estimate]**.	
11	Read the GPU workload comparison. Expected B300 = `16`, B200 = `16`, H200 = `24` GPUs. These workload counts are separate from the `256` GPU factory deployment anchor. Click **[Export result]** to save the calculation.	

Step	Activity and exact values	Screen reference
12	Click **[Continue to Design handoff]**, then **[Prepare projection]**. Read the executive summary and download the design intake and companion specification. Unresolved network and storage measurements remain explicit.	

Step	Activity and exact values	Screen reference
13	Click **[Explain this to me]**. Set **Explanation connection** = `Deterministic explanation — no model`. Click **[Generate explanation]**. The saved calculation cannot be changed by this action.	

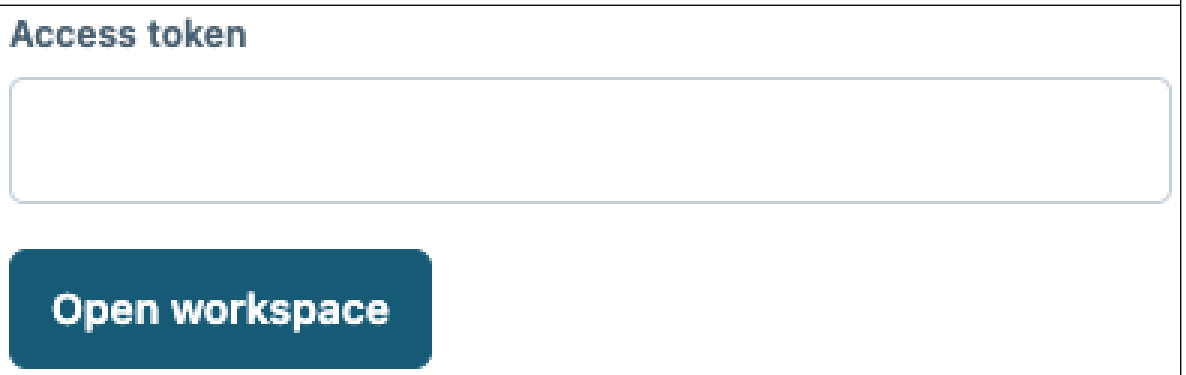
Step	Activity and exact values	Screen reference
14	Read the explanation of the workload GPUs, deployment anchor, memory, storage and all network roles. Expand **[Source fields and assumptions]** to trace the explanation. Click **[Download explanation]**, then **[Close]**.	

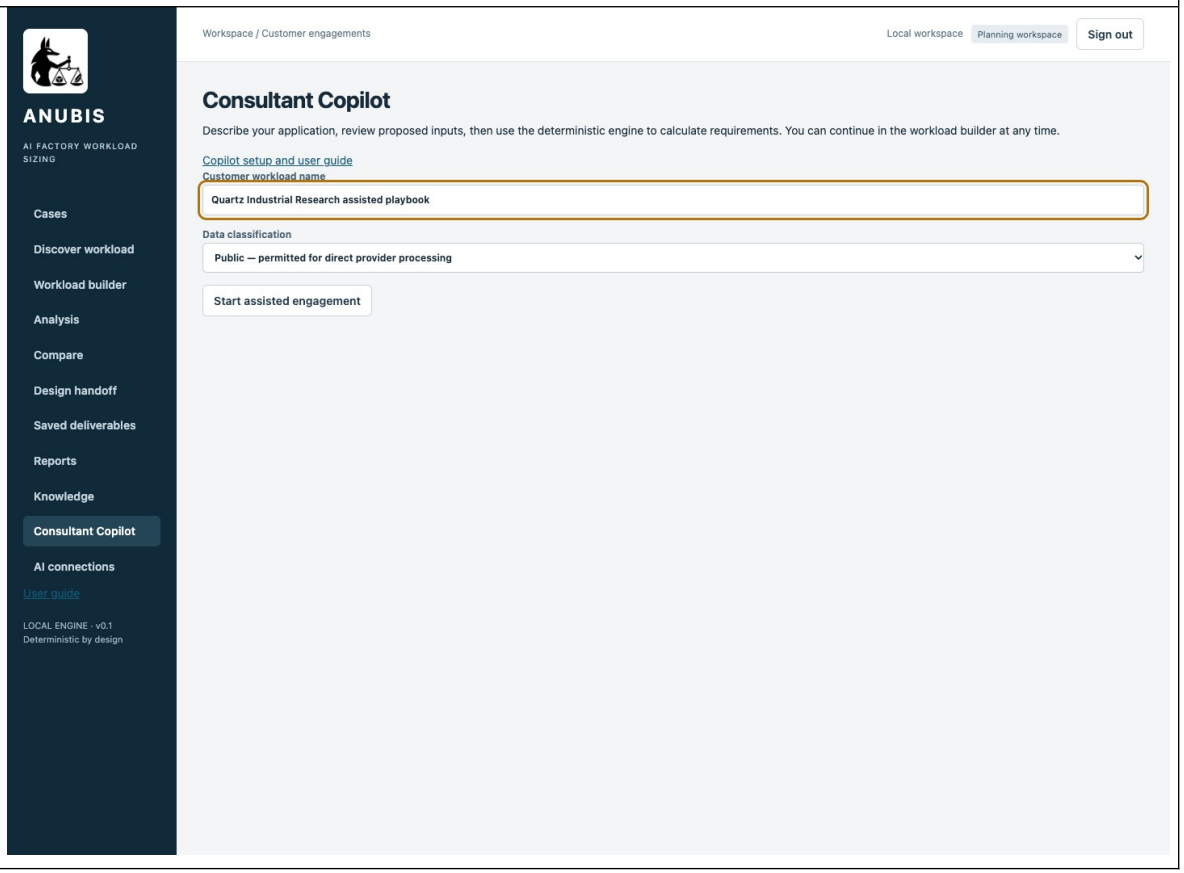
Step	Activity and exact values	Screen reference
15	Scroll down within **[Explain this to me]** to read **[Factory networks]** and **[What to do next]**. Expand **[Source fields and assumptions]** if needed. Click **[Download explanation]** to save the text. Scroll to the top and click **[Close]** to continue.	

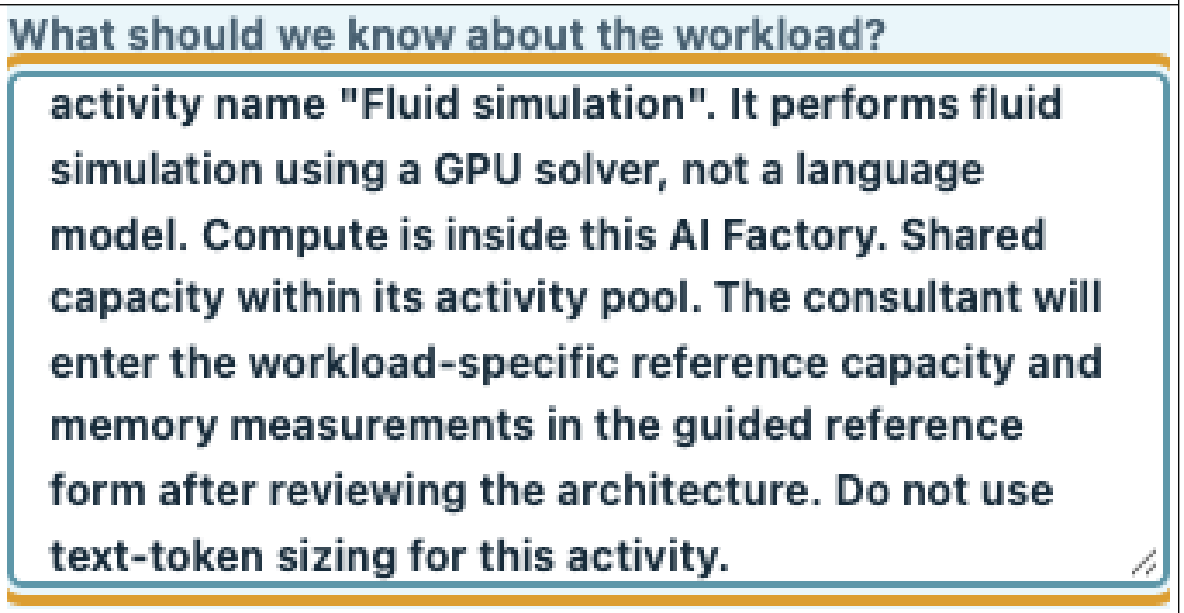
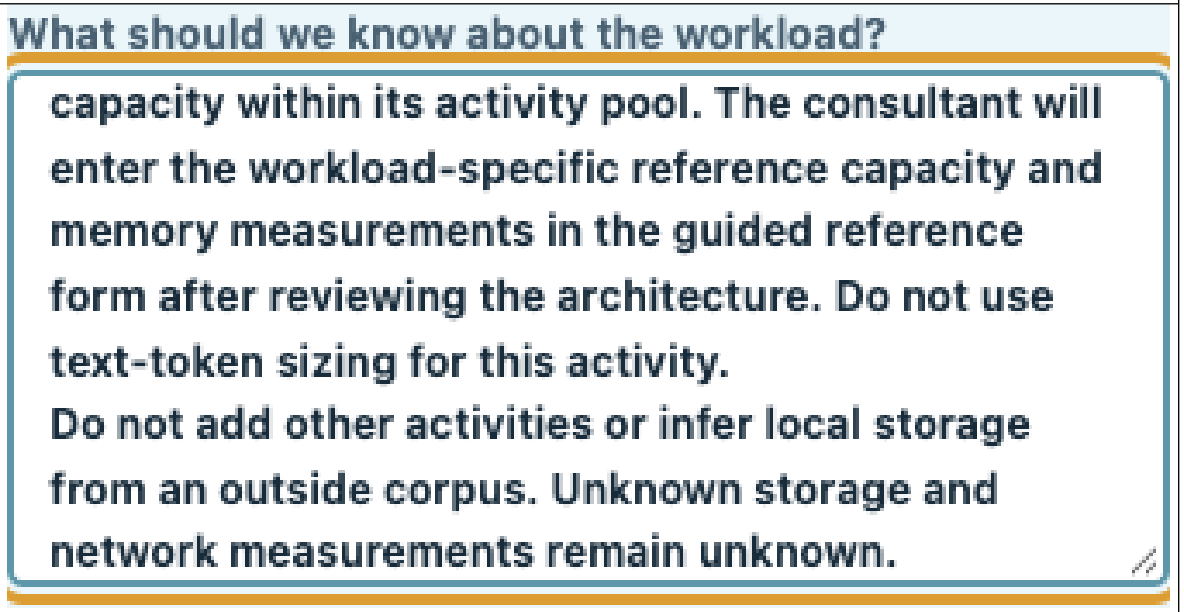
Step	Activity and exact values	Screen reference
16	Set **Completion notes** = `Planning requirements and explanation delivered. Outstanding network and storage measurements remain in the companion specification.`. Check **I confirm that these selected requirements are delivered.**. Click **[Complete engagement]**.	
17	The engagement is complete. Its exact calculation, handoff and explanation remain saved. Use **[Cases]** to reopen the engagement and **[Saved deliverables]** to retrieve its documents.	

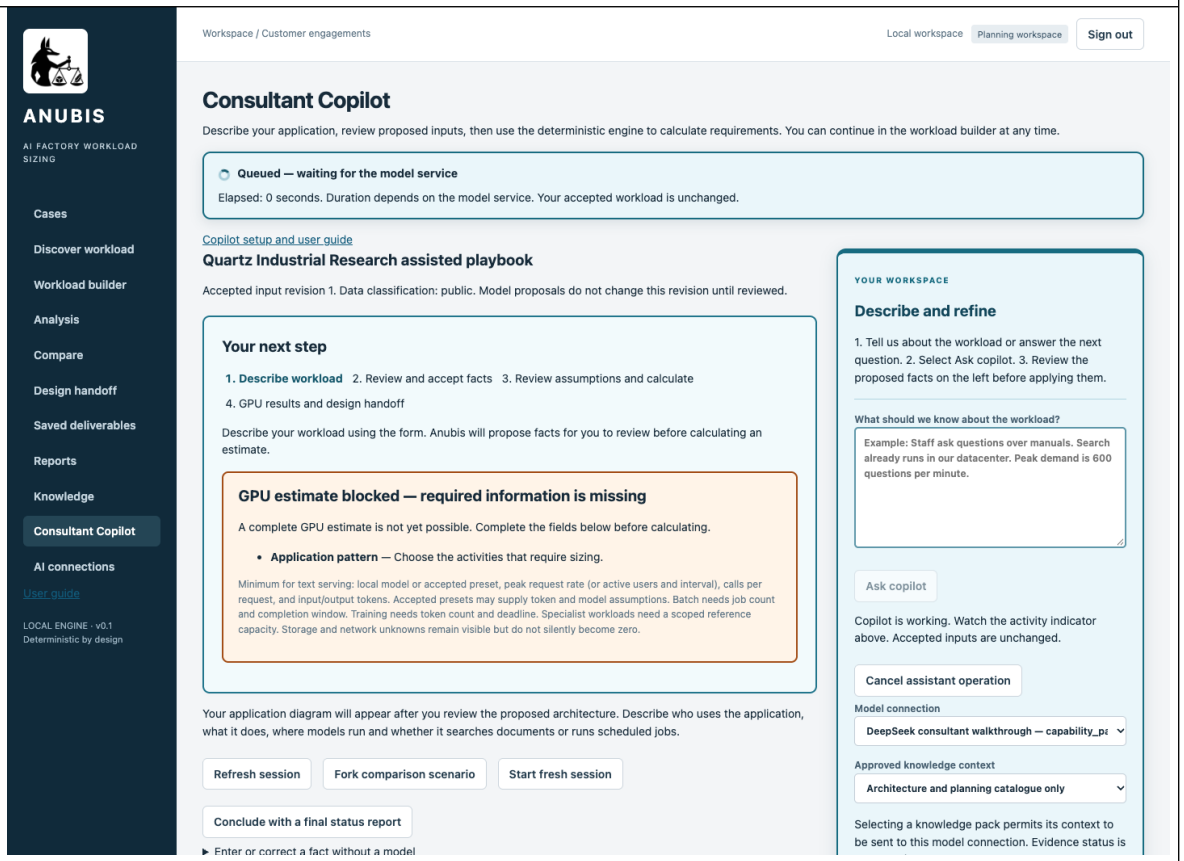
Copilot assisted workflow

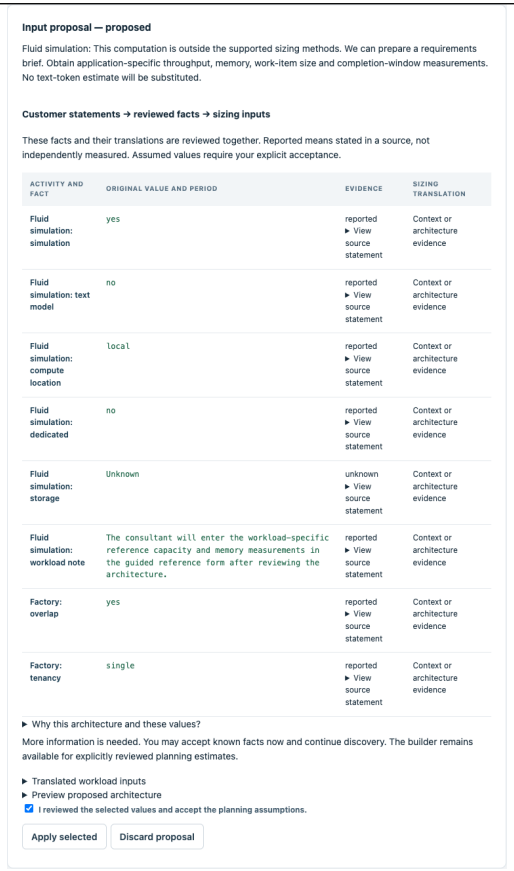
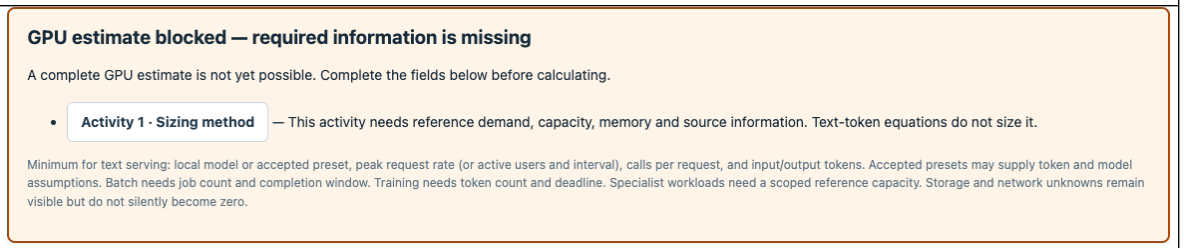
Follow these numbered steps in order. Use a separate engagement name for this route. Input values are the same workload assumptions in both routes. In the assisted route, inspect extracted facts and explicitly enter any specialist reference values in the guided form.

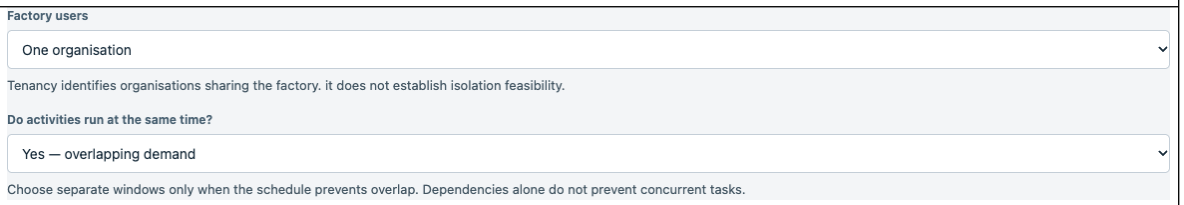
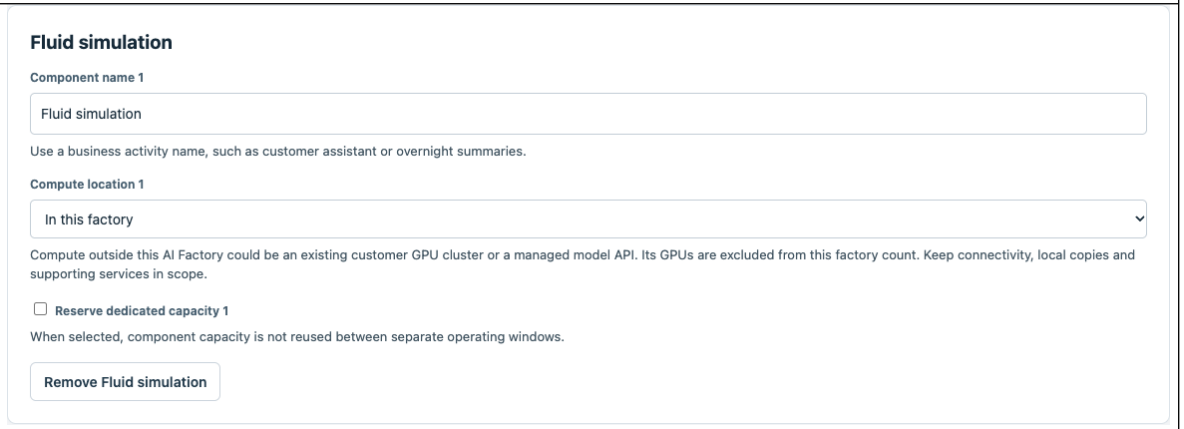
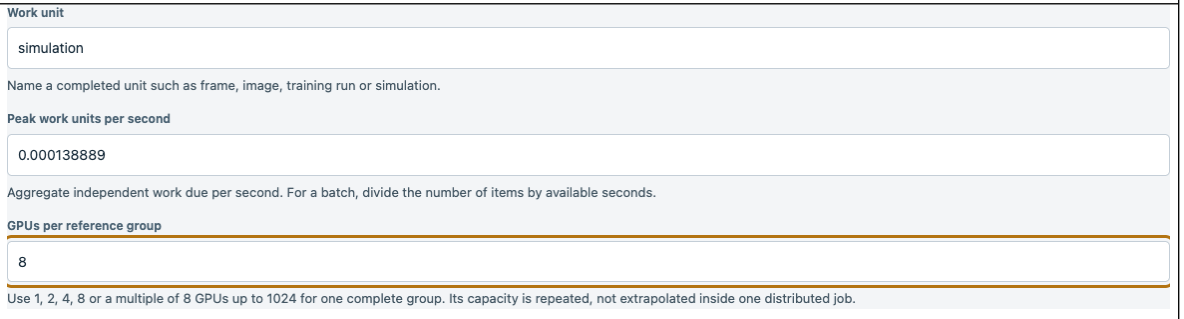
Step	Activity and exact values	Screen reference
1	Start Anubis and open the local address. Retrieve your private token from `.anubis/identities.json`. Paste it into **Access token** and click **[Open workspace]**. Never share the token.	

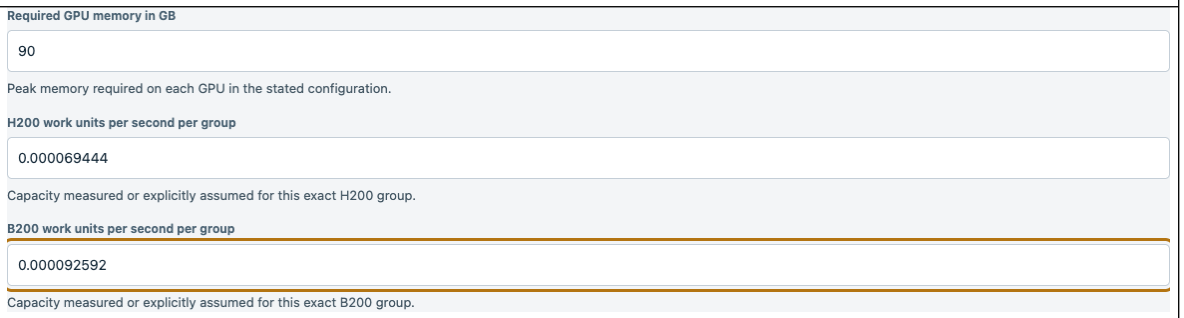
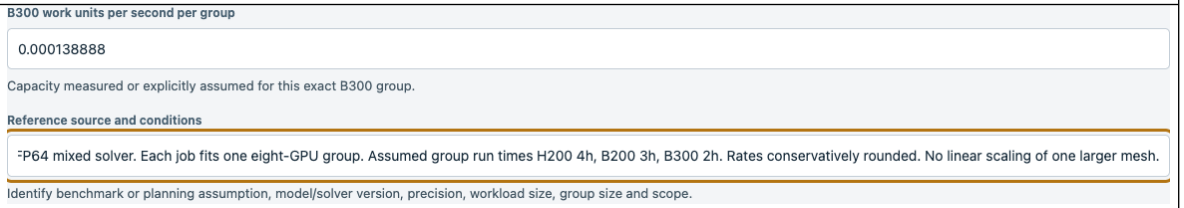
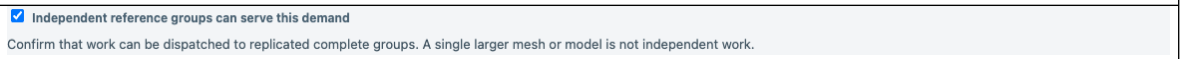
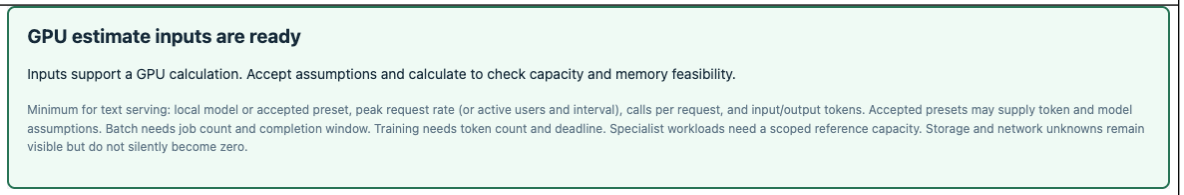
Step	Activity and exact values	Screen reference
2	Click **[Consultant Copilot]**. Set **Customer workload name** = `Quartz Industrial Research assisted playbook`. Set **Data classification** = `Public — permitted for direct provider processing`. Click **[Start assisted engagement]**.	

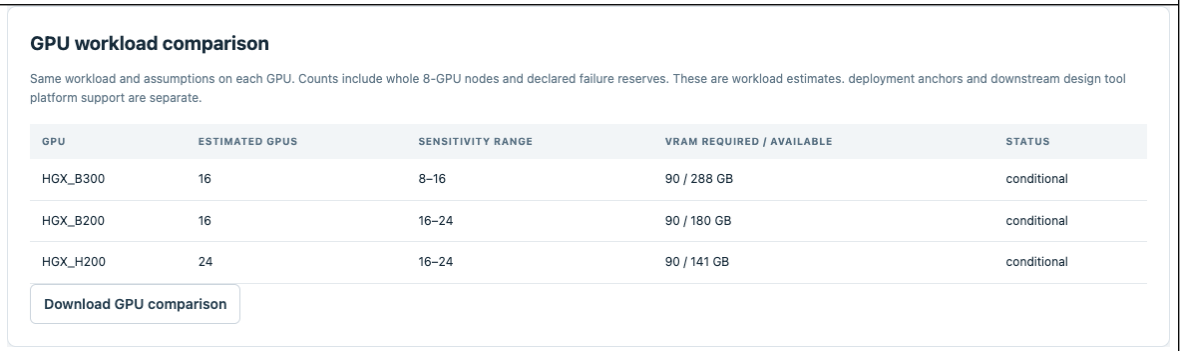
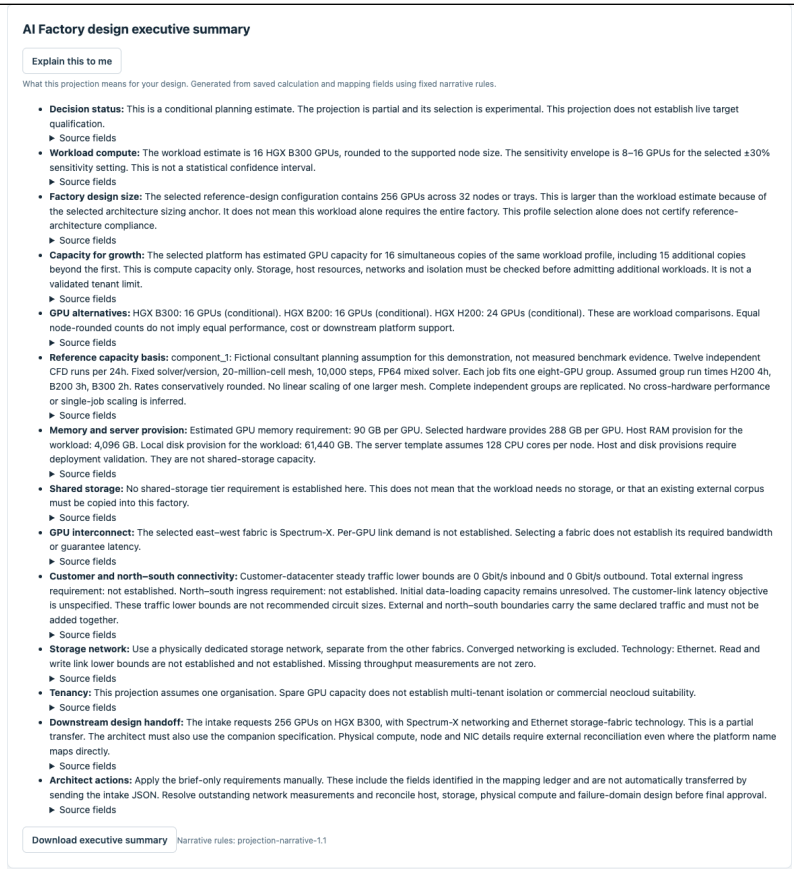
Step	Activity and exact values	Screen reference
3	Select your tested connection in **Model connection**. Enter the following paragraph in **What should we know about the workload?**: `Fictional customer Quartz Industrial Research. This is a planning exercise. Activities can overlap. A single organisation uses the factory. Use the exact activity name "Fluid simulation". It performs fluid simulation using a GPU solver, not a language model. Compute is inside this AI Factory. Shared capacity within its activity pool. The consultant will enter the workload-specific reference capacity and memory measurements in the guided reference form after reviewing the architecture. Do not use text-token sizing for this activity..`	
4	Append the following paragraph to **What should we know about the workload?**: `Do not add other activities or infer local storage from an outside corpus. Unknown storage and network measurements remain unknown..` Click **[Ask copilot]**.	

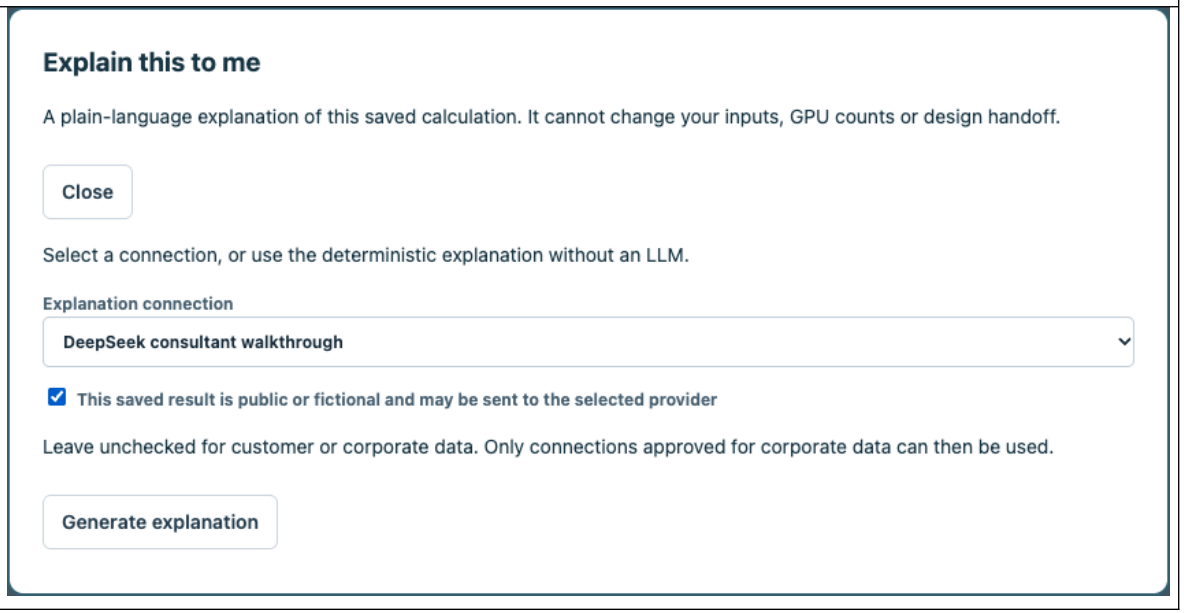
Step	Activity and exact values	Screen reference
5	Wait while Copilot interprets the description. The active status indicator confirms that it is still working. Do not submit duplicate requests.	

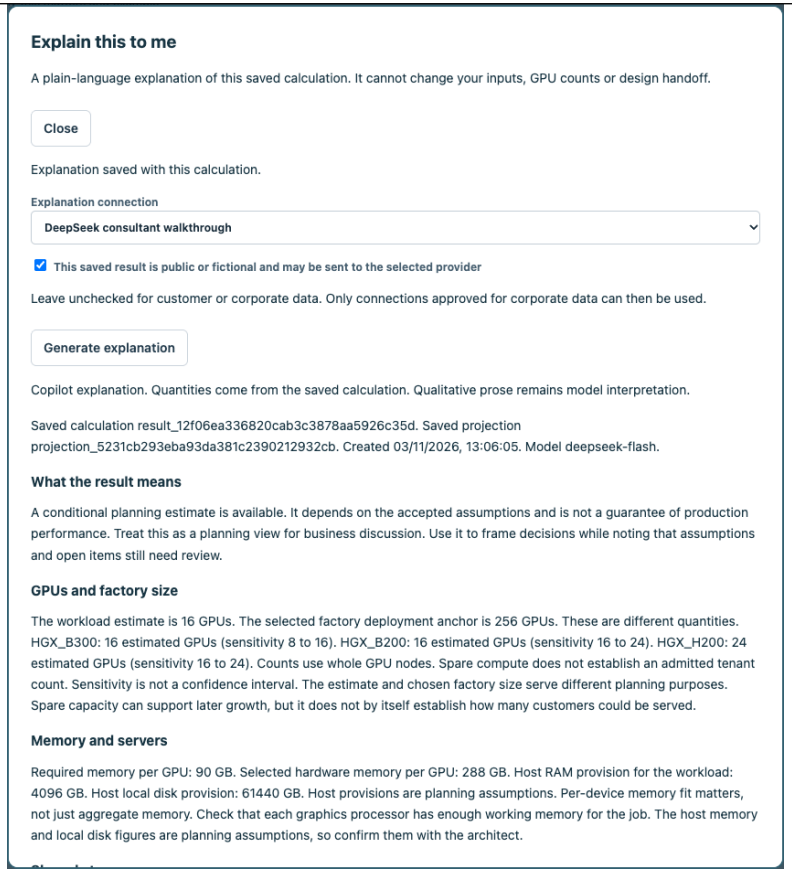
Step	Activity and exact values	Screen reference
6	Review the proposed activities, placement, measurements and assumptions against the input register. Check **I reviewed the selected values and accept the planning assumptions.**. Click **[Apply selected]**. Specialist reference inputs are entered during the next review.	
7	Click **[Review and calculate GPU estimate]**. The readiness panel names any missing inputs. Complete them in the guided review. Copilot has not invented reference throughput.	

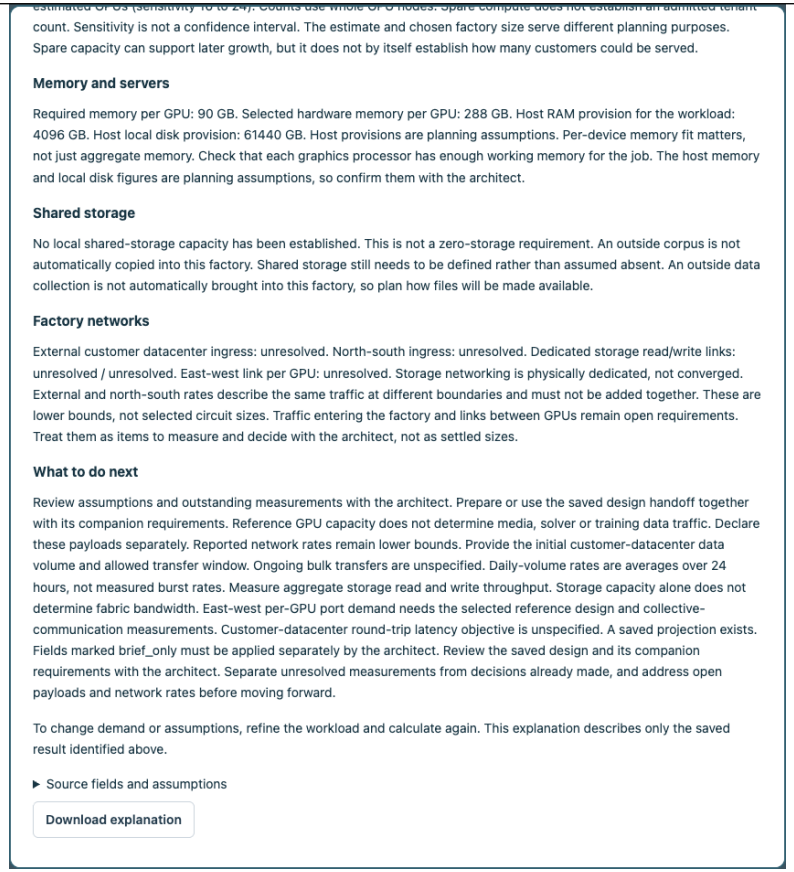
Step	Activity and exact values	Screen reference
8	Set **Factory users** = `One organisation`. Set **Do activities run at the same time?** = `Yes — overlapping demand`.	
9	Set **Component name 1** = `Fluid simulation`. Set **Compute location 1** = `In this factory`. Set **Reserve dedicated capacity 1** ***unchecked***.	
10	Select **[Fluid simulation]**. Set **Sizing method** = `Reference capacity for independent work`. This uses complete independent GPU groups and the explicit scenario assumptions.	
11	Set **Work unit** = `simulation`. Set **Peak work units per second** = `0.000138889`. Set **GPUs per reference group** = `8`.	

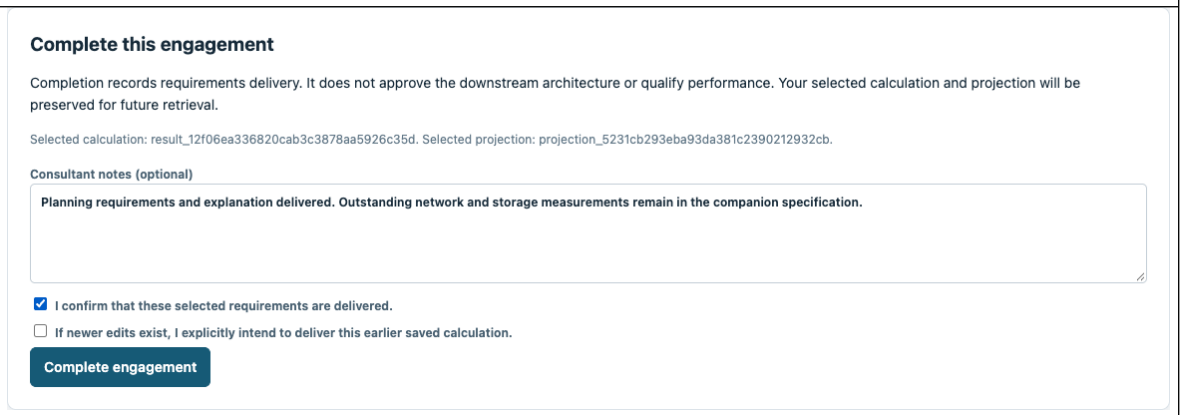
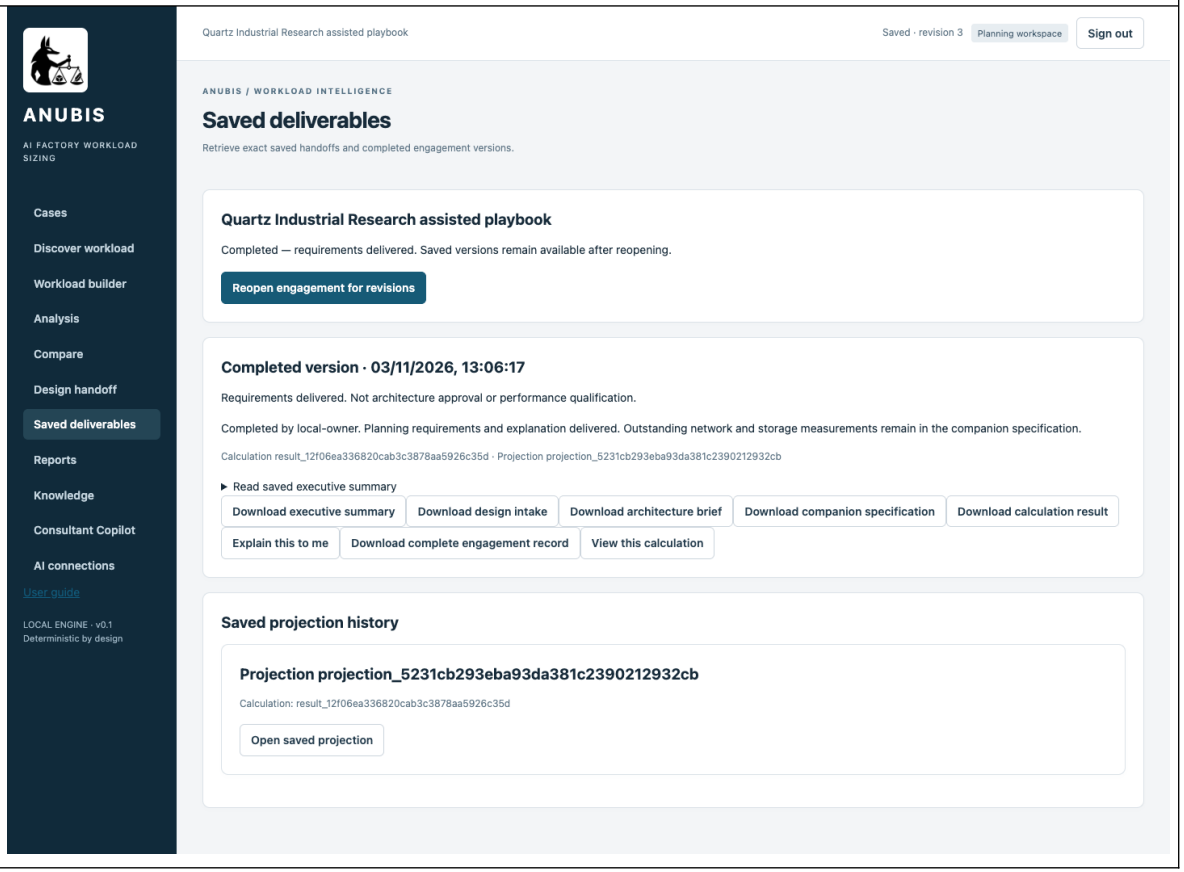
Step	Activity and exact values	Screen reference
12	Set **Required GPU memory in GB** = `90`. Set **H200 work units per second per group** = `0.000069444`. Set **B200 work units per second per group** = `0.000092592`.	
13	Set **B300 work units per second per group** = `0.000138888`. Set **Reference source and conditions** = `Fictional consultant planning assumption for this demonstration, not measured benchmark evidence. Twelve independent CFD runs per 24h. Fixed solver/version, 20-million-cell mesh, 10,000 steps, FP64 mixed solver. Each job fits one eight-GPU group. Assumed group run times H200 4h, B200 3h, B300 2h. Rates conservatively rounded. No linear scaling of one larger mesh..`	
14	Check **Independent reference groups can serve this demand** and **I accept this reference capacity and its scope**. Leave **Reference includes local search and indexing *unchecked*** for this component. These example capacities are accepted planning assumptions, not measured benchmark results.	
15	Confirm **[GPU estimate inputs are ready]**. Retain the displayed B300 enterprise deployment profile and **Sensitivity envelope** = `±30%`. Check **Use these planning assumptions**. Click **[Calculate estimate]**.	

Step	Activity and exact values	Screen reference
16	Read the GPU workload comparison. Expected B300 = 16, B200 = 16, H200 = 24 GPUs. These workload counts are separate from the 256 GPU factory deployment anchor. Click **[Export result]** to save the calculation.	
17	Click **[Continue to Design handoff]**, then **[Prepare projection]**. Read the executive summary and download the design intake and companion specification. Unresolved network and storage measurements remain explicit.	

Step	Activity and exact values	Screen reference
18	Click **[Explain this to me]**. Set **Explanation connection** = `DeepSeek consultant walkthrough`. Check **This saved result is public or fictional and may be sent to the selected provider** for this fictional exercise. Click **[Generate explanation]**. The saved calculation cannot be changed by this action.	

Step	Activity and exact values	Screen reference
19	Read the explanation of the workload GPUs, deployment anchor, memory, storage and all network roles. Expand **[Source fields and assumptions]** to trace the explanation. Click **[Download explanation]**, then **[Close]**.	

Step	Activity and exact values	Screen reference
20	Scroll down within **[Explain this to me]** to read **[Factory networks]** and **[What to do next]**. Expand **[Source fields and assumptions]** if needed. Click **[Download explanation]** to save the text. Scroll to the top and click **[Close]** to continue.	

Step	Activity and exact values	Screen reference
21	Set **Completion notes** = `Planning requirements and explanation delivered. Outstanding network and storage measurements remain in the companion specification.`. Check **I confirm that these selected requirements are delivered.**. Click **[Complete engagement]**.	
22	The engagement is complete. Its exact calculation, handoff and explanation remain saved. Use **[Cases]** to reopen the engagement and **[Saved deliverables]** to retrieve its documents.	

Compare and retrieve the results

Check	Manual route	Copilot assisted route
Workload GPU counts	B300 16, B200 16, H200 24	B300 16, B200 16, H200 24
Design handoff	Prepared with companion requirements	Prepared with companion requirements
Explanation	Saved deterministic narrative	Saved model-assisted narrative
Completion	Requirements delivered and saved	Requirements delivered and saved

Open the engagement from Cases and select Saved deliverables. Download the saved calculation, intake, brief and companion specification. Explain this to me retrieves narratives associated with that exact result. A later calculation does not update an earlier explanation. Refining demand requires a fresh deterministic calculation.

Appendix Parameter collection guide

ID	Field and meaning	How to obtain or decide it	When unavailable
I01	**Customer workload name** Identifies the two separate saved runs.	Choose the route suffix to avoid mixing results.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I02	**Factory users** Organisations sharing the factory.	Confirm commercial and organisational scope. Tenancy does not multiply aggregate demand automatically.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I03	**Do activities run at the same time?** Peak workloads may coincide.	Review schedules and telemetry with the operations owner.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I04	**Component name** Business activity being sized. Numbered labels follow the order of components in the active route.	Agree a recognisable activity name with the application owner. Use the matching diagram to check its scope.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I05	**Application pattern** Application behaviour, not a server type.	Walk through a customer task. Identify text serving, retrieval, media processing or solver work.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I06	**Compute location** Where computation runs relative to this factory.	Confirm the hosting owner and deployment boundary. Outside services do not contribute local model GPUs.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I07	**Relationship** Order or relationship of activities. This is separate from whether demand overlaps.	Review the business flow with the application owner. Confirm scheduling separately.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I08	**Sizing method** Analytical text sizing or replication of a complete reference GPU group.	Use analytical sizing for supported language models. Select reference capacity for a scoped specialist workload.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.

ID	Field and meaning	How to obtain or decide it	When unavailable
I09	**Peak work units per second** Aggregate independent work units due per second.	Count peak arriving work. For batch work divide items by available seconds. Camera demand can be cameras multiplied by sampled frames per second.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I10	**Work unit** Completed unit of work, such as a processed frame or complete simulation.	Define the exact work item, input dimensions, quality settings, precision and success condition with the application owner.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I11	**GPUs per reference group** GPUs in a complete independently repeatable group.	Read the tested or assumed deployment configuration. Use one, two, four, eight or a multiple of eight GPUs. Do not scale a single larger mesh by adding unrelated groups.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I12	**Required GPU memory in GB** Peak memory required on each GPU in the reference configuration.	Measure the busiest device with GPU telemetry during representative work. Include weights, working state and buffers. Record the source and margin.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I13	**H200 work units per second per group** Capacity per complete group on this specific GPU platform, in the same work units per second as demand.	Use a matching measured run or an explicit engineering assumption for this platform. Do not silently translate throughput measured on another GPU.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I14	**B200 work units per second per group** Capacity per complete group on this specific GPU platform, in the same work units per second as demand.	Use a matching measured run or an explicit engineering assumption for this platform. Do not silently translate throughput measured on another GPU.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I15	**B300 work units per second per group** Capacity per complete group on this specific GPU platform, in the same work units per second as demand.	Use a matching measured run or an explicit engineering assumption for this platform. Do not silently translate throughput measured on another GPU.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I16	**Independent reference groups can serve this demand** Whether independent full groups can share the incoming work.	Check that the scheduler can route separate jobs or requests to each group. A single job requiring a larger distributed system does not satisfy this assumption.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I17	**I accept this reference capacity and its scope** Consultant acceptance of reference scope and assumptions.	Review demand, capacity, memory and group independence together. Reaccept after Copilot refinement only if the reference remains applicable.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.

ID	Field and meaning	How to obtain or decide it	When unavailable
I18	**Reference source and conditions** Source and operating conditions for the capacity assumption.	Record benchmark or engineering-assumption provenance, date, model or solver version, precision, input dimensions, group size and scope. Fictional example values are not measurements.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I19	**Infrastructure deployment profile** Selects factory anchors rather than workload demand.	Architect selects a compatible catalogue profile.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I20	**Sensitivity envelope** Explores faster and slower throughput assumptions.	Choose the agreed planning sensitivity. It is not a confidence interval.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I21	**Use these planning assumptions** Accepts the displayed planning defaults.	Review defaults before calculating.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I22	**Locally retained data in TB / Daily growth in GB** No local shared-storage amount is established. Blank is not zero.	Inventory actual local retained copies. Do not copy an outside corpus into this value.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I23	**Initial dataset to load in TB / Initial loading window in hours** Initial transfer scope and time remain unresolved.	Obtain transfer volume and the allowed window from the data owner.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I24	**Ongoing incoming / outgoing data in GB per day** Bulk daily transfer volumes remain unspecified.	Measure ingress and egress counters separately from model token traffic.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I25	**Aggregate storage reads / writes in Gbps** Dedicated storage fabric throughput remains unresolved.	Measure directional read and write payload peaks. Capacity alone cannot determine bandwidth.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I26	**East-west payload per GPU in Gbps / Customer datacenter round-trip target in ms** Collective traffic and latency objective remain unresolved.	Use GPU communication telemetry and an agreed RTT objective.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.

ID	Field and meaning	How to obtain or decide it	When unavailable
I27	**Usable link fraction / Initial loading overlap** Accepted network defaults. Storage networking is physically dedicated.	Review usable bandwidth and whether loading coincides with normal operation.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I28	**Explanation connection** Chooses prose generation without changing the calculation.	Configure an approved provider in AI connections. Backend credentials stay on the server.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.
I29	**This saved result is public or fictional and may be sent to the selected provider** Permission to send the saved explanation packet to the chosen connection.	Leave unchecked for corporate data and select a connection approved for that data.	Record the unknown or an explicitly accepted assumption. Resolve any field listed by GPU estimate blocked before calculating.

Appendix Placement and explanation boundaries

Term	Meaning and concrete examples
Outside this AI Factory	Compute, search or storage beyond the capacity being designed. Examples include a customer GPU cluster, managed model API, customer datacenter search cluster, cloud search platform or an existing object store. It does not necessarily mean public Internet or a different site.
Source corpus and index	Original documents and the searchable index have separate storage and compute footprints. Examples include an NFS or SMB share, object storage, SharePoint and an OpenSearch index. Outside data is not automatically a local retained copy or initial transfer volume.
Model explanation	The LLM explains a saved result using a read-only packet. Numeric facts are inserted by the server. Invalid or unavailable model responses fall back to deterministic text. Qualitative prose remains interpretation, not a new engineering decision.
Reference capacity	Specialist counts use the exact per-hardware throughput and memory assumptions in the register. The example values are fictional planning assumptions. A complete independent group can be replicated. No linear scaling of a single larger job is inferred.
Validation scope	Both GUI routes were executed against the application and checked for matching GPU comparisons and saved handoffs. This validates the software workflow. It does not qualify real hardware performance, model quality, latency or live-provider reliability for every future request.